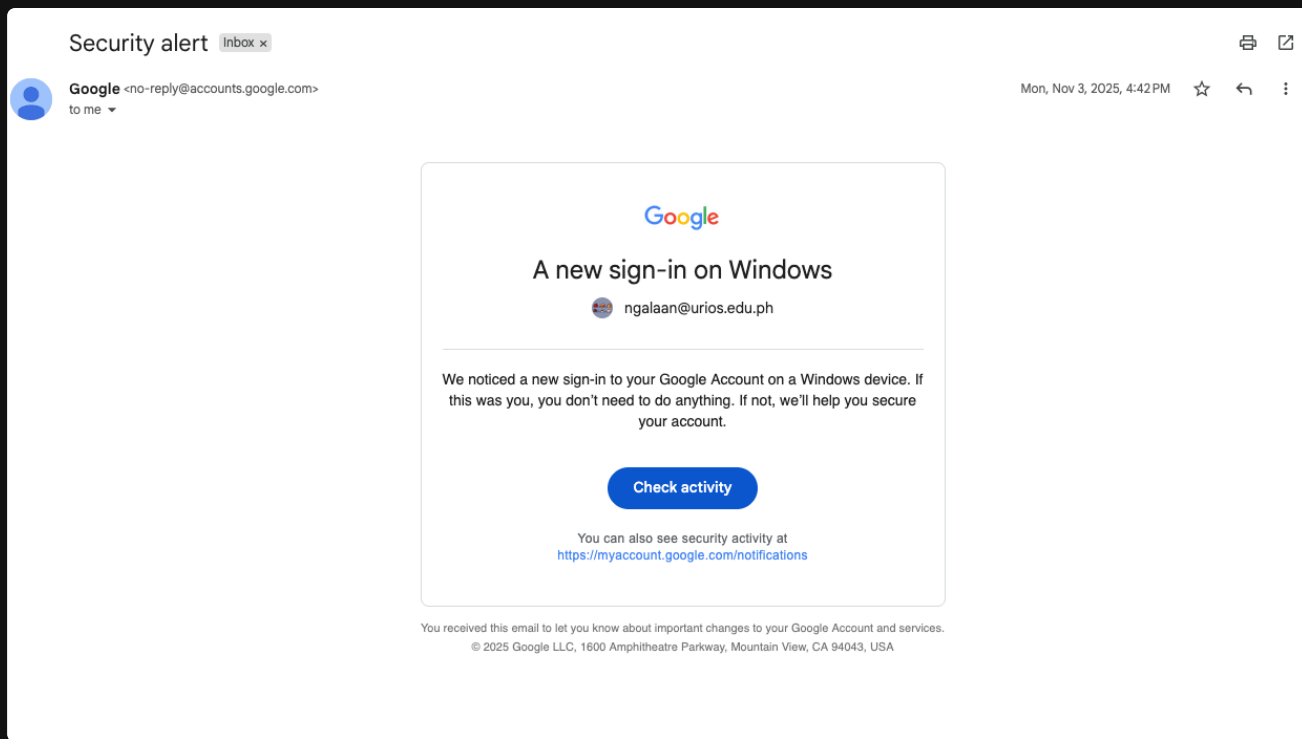
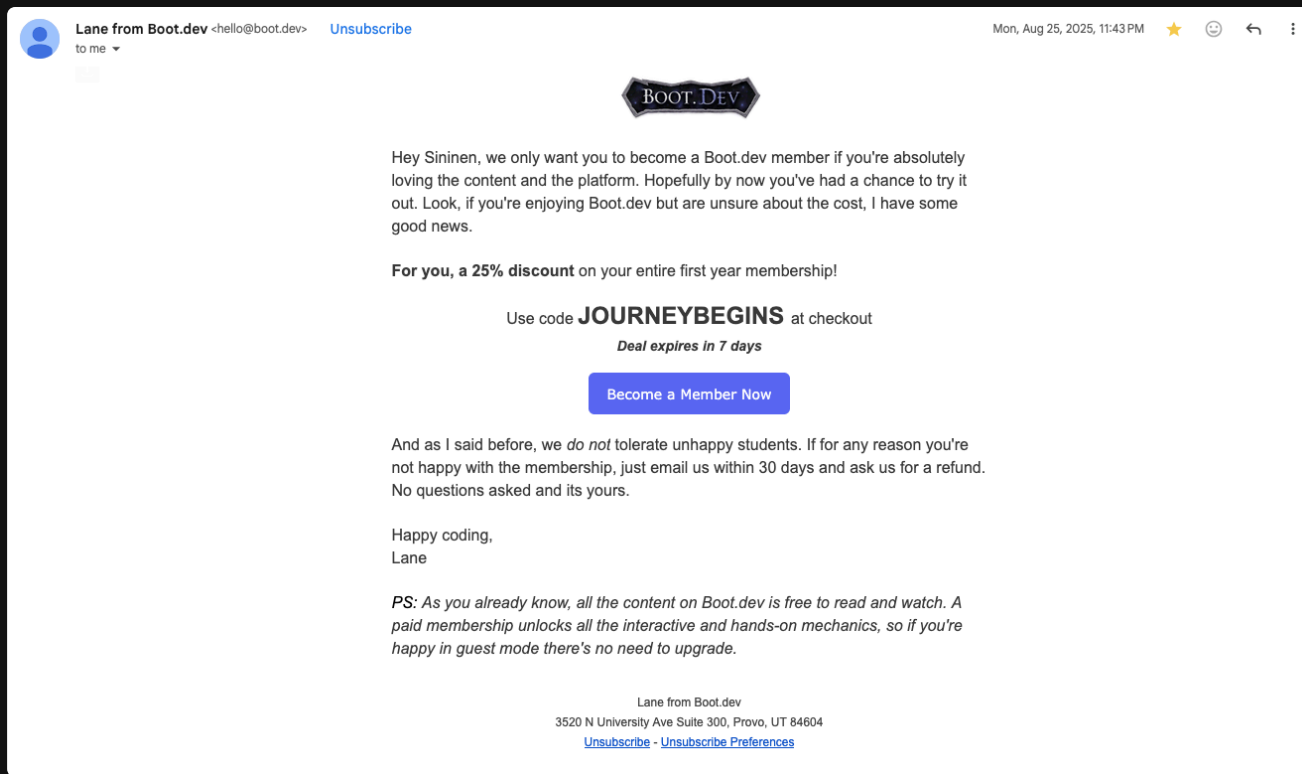


Digital threats and cybercrimes

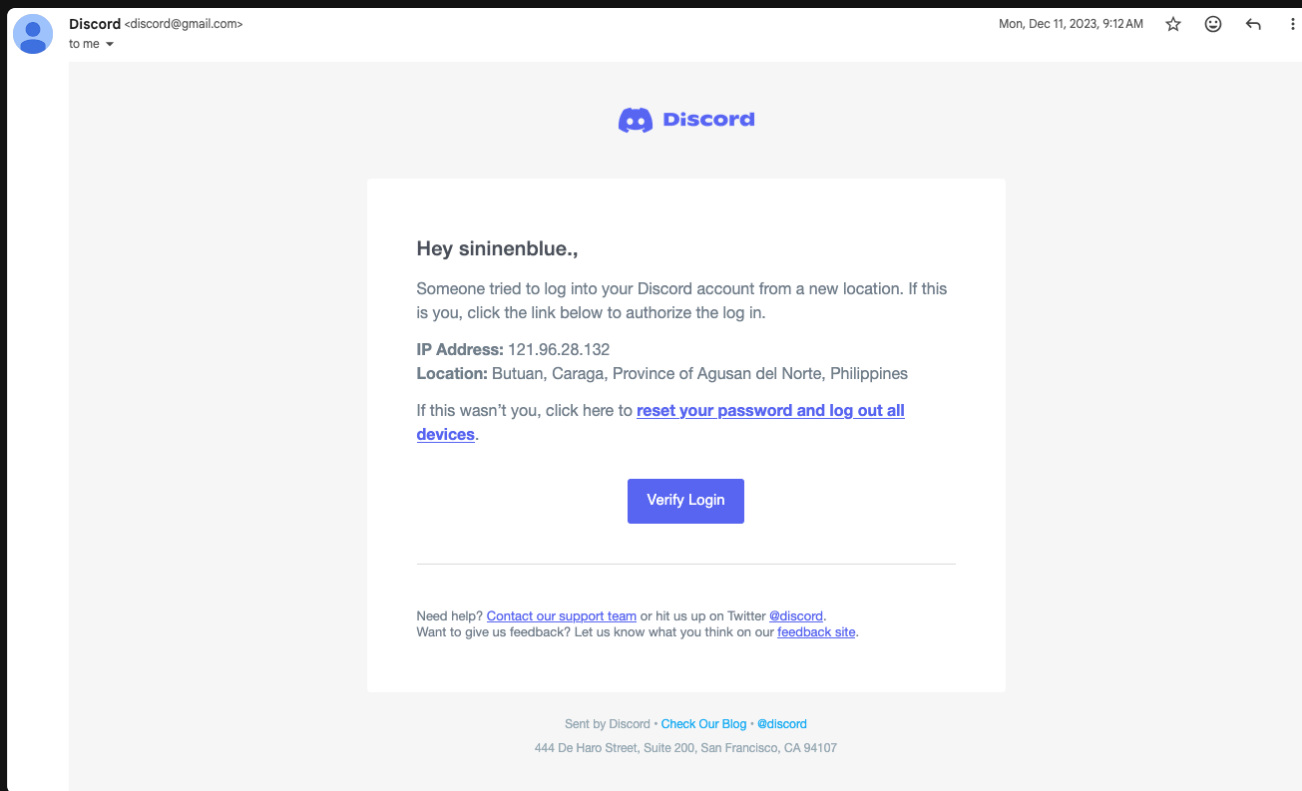
Would you click on this link?



Would you click on this link?



Would you click on this link?



Digital Threats

Types of threats in the digital world

There is *technically* only one type of threat purely in the digital world, and that is **malware**

However, there are many other types of threats that are **aided** by the digital world

These threats are all methods of **compromising** a person's device, data, or identity

And regardless of if they are successful, are considered **cybercrimes** such as:

- data breaches
- financial fraud
- identity theft

Malware

Malware or **malicious software** is any software that is designed to cause cause your computer, server, client, or network to behave in an *unwanted* manner

This means

1. slowing your computer down
2. showing you ads
3. causing system crashes
4. stealing your data

In the modern day, malware is mostly used for *financial* gain,

- Either by **stealing data** that can be sold,
- Holding your data **hostage** until you pay a ransom,
- Or by using your computer to **mine cryptocurrency** for the attacker

Types of Malware

Types of Malware

There are 2 aspects to malware, the **method of infection** and the **payload**, how you get it and what it does

There are 2 main methods of **infection**:

1. A *lie*

This can come in the form of an email, a message, a pop-up, an ad, a download link, etc. You are *tricked* into clicking something where you expect something else to happen

2. A *vulnerability*

This is when the malware exploits a **vulnerability** in your software, such as an unpatched security flaw, to infect your device without you having to click on anything.

This is usually done through *worms* and *viruses* that can spread themselves across networks and devices.

Types of Malware

In terms of **payload**, there are many types of malware, however all have the same goal of **compromising** your device, data, or identity

Usually for the purpose of financial gain, but sometimes just for fun or to cause harm

Starting from the least common (at least for the average user) to the *most common*, we have:

1. *Ransomware*: encrypts your files and demands a ransom to unlock them
2. *Spyware*: secretly monitors your activities and steals your data
3. *Adware*: shows you unwanted ads and generates revenue for the attacker
4. *Botnet malware*: allows your computer to be controlled to perform malicious activities,
 - such as sending spam emails,
 - launching DDoS attacks,
 - or mining cryptocurrency

Threats aided by the digital world

Many cybercrime threats are not **exclusive** to the digital world, but are **aided** by it, such as:

- Social engineering
- Identity theft
- Harassment
- Scams

Once again, these are all methods of **compromising** a person's device, data, or identity.

And while possible **without** the internet, the digital world has made these threats much more **prevalent** and **dangerous**

Especially with the rise of **social media** and **AI**, where it's now possible to create **deepfakes** and **fake profiles** that can be used for social engineering, harassment, and scams

Laws

Cybercrime prevention act of 2012 (Republic Act No. 10175)

AN ACT DEFINING CYBERCRIME, PROVIDING FOR THE PREVENTION, INVESTIGATION, SUPPRESSION AND THE IMPOSITION OF PENALTIES THEREFOR AND FOR OTHER PURPOSES

Laws

Punishable offenses include:

- *against confidentiality*

- illegal access
- interference
- misuse of devices
- cyber-squatting

- *using computers*

- forgery
- fraud
- identity theft

- *against content*

- cybersex
- child pornography
- unsolicited commercial communications
- libel

Laws

Penalties include

- at least two hundred thousand pesos (200,000.00) in fines
- up to five million pesos (5,000,000.00)
- prison mayor (6 years and 1 day to 12 years)

How to protect yourself from these threats

Personally:

- Slow down

The most common way people get hacked, scammed, or otherwise compromised is by *clicking on a link* or *downloading a file* that they shouldn't have

Whenever someone sends you a link, a file, especially if it's from *someone you don't know*, or if it's unexpected, take an extra moment to **think** before you click

Cybercrimes, especially scams, **filter** out the most vulnerable targets

You might not get caught by obvious scams, but that leaves you confident

How to protect yourself from these threats

On the software side:

- stay up to date: your operating system, your browser, and other apps have *security updates* that fix vulnerabilities, so make sure to install them as soon as they are available
- Strong passwords:
 - ideally use a *password manager* to generate and store strong, unique passwords for each of your accounts
 - otherwise, make them **long, mixing** letters, numbers, and symbols
 - and **don't** use the same password, at least try to make them different
 - I recommend using a **passphrase** instead of a password, which is a sequence of random words that is easy to remember but hard to guess
- Use Two-factor authentication